ia_study2_mainAnalyses

Verity Y. Q. Lua

2026-03-12

```r
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
library(emmeans)
library(psych)
library(patchwork)
source("https://raw.githubusercontent.com/vyqlua/avi_functions/refs/heads/main/ipsatize_fn.R")
```

Load Data

```
data <- read_csv("ia_study2_anon_data_computed.csv")
data <- data[ , -1]

levelBreakdown <- read_csv("ia_TrialsBreakdown2.csv")
```

Are There Cultural Differences in Ideal Affect between European Americans and Taiwanese?

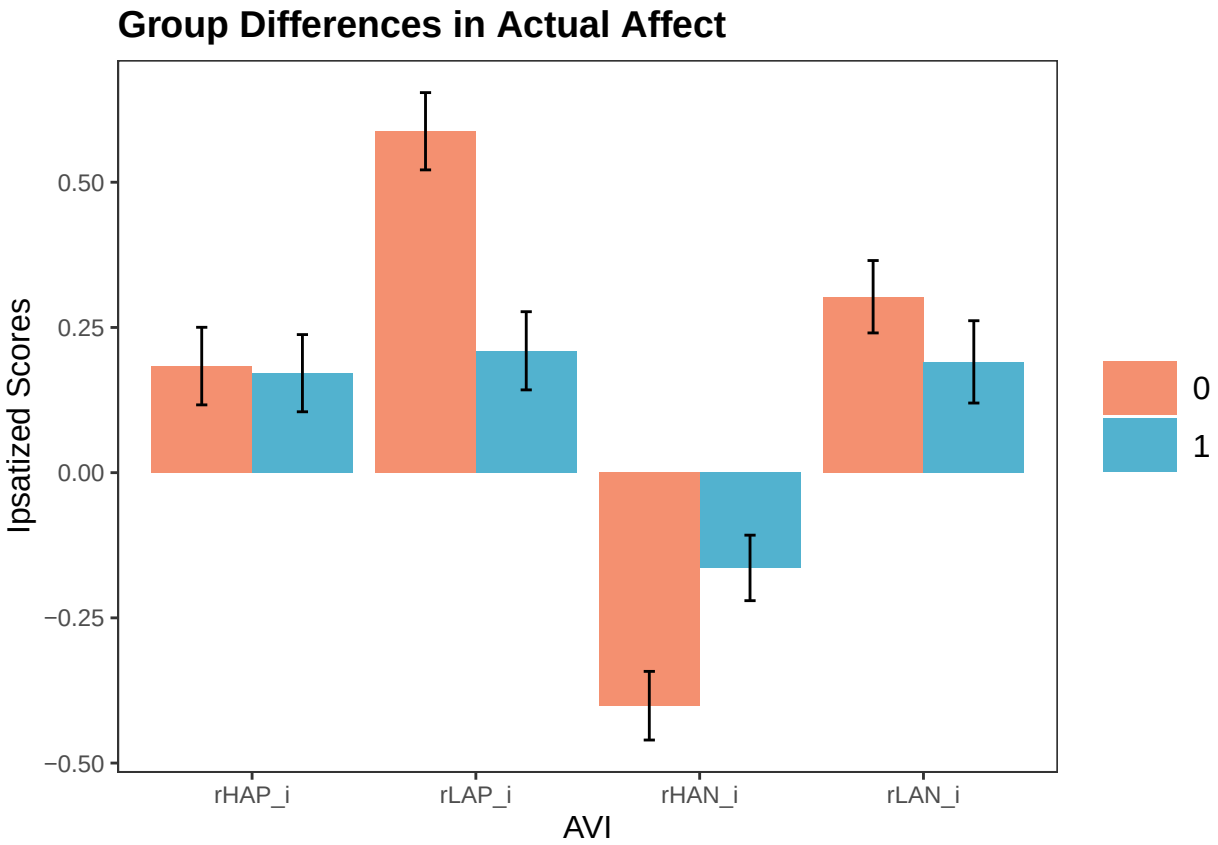
Visualization

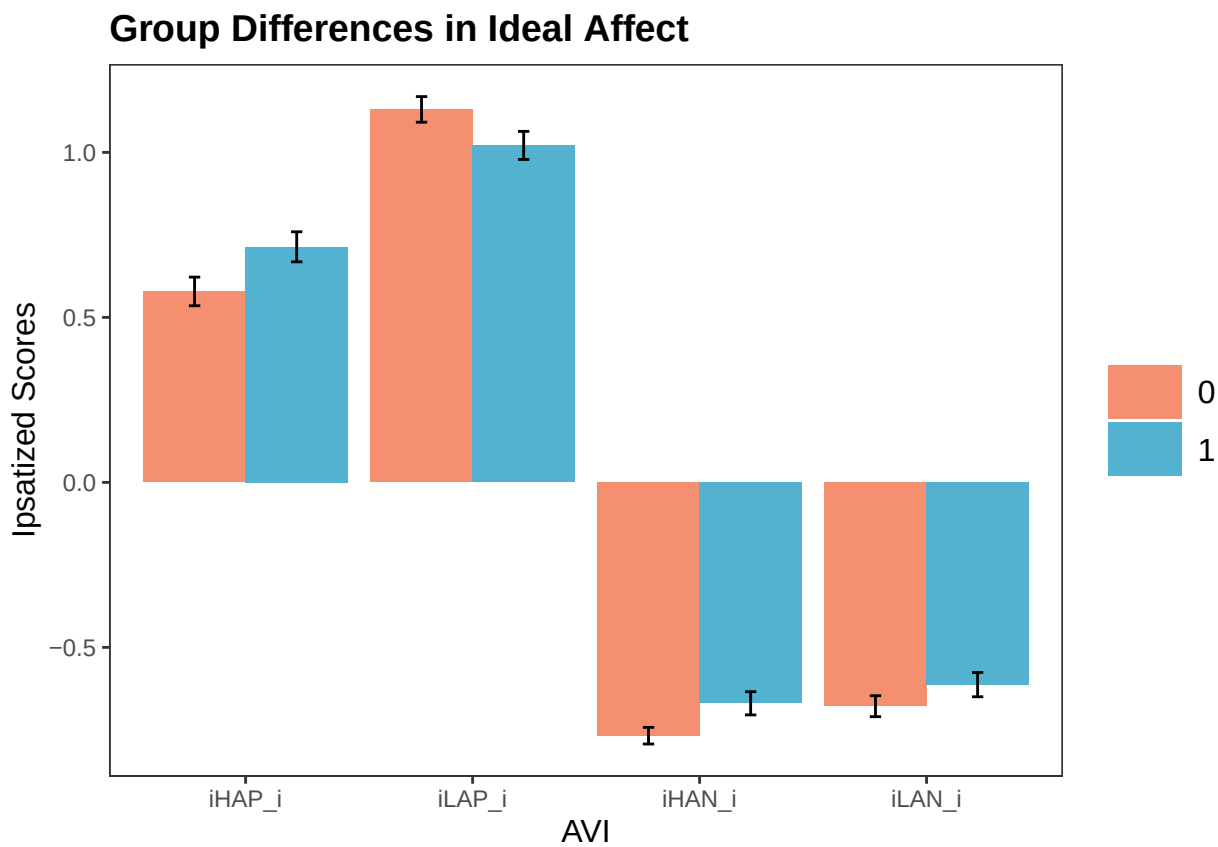
```r
plot_avi(data, group_id = "cultureEA")
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use ‘all_of()‘ or ‘any_of()‘ instead.
##   # Was:
##   data %>% select(group_id)
##
##   # Now:
##   data %>% select(all_of(group_id))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
## ‘summarise()‘ has grouped output by ’cultureEA’. You can override using the
## ‘.groups‘ argument.
```

```
## # A tibble: 8 x 6
## # Groups:   cultureEA [2]
##   cultureEA AVI      mean    sd     se AVI_order
##   <fct>     <fct>   <dbl> <dbl>  <dbl>     <dbl>
## 1 0         rHAP_i  0.183 0.688 0.0668         1
## 2 1         rHAP_i  0.171 0.716 0.0665         1
## 3 0         rLAP_i  0.588 0.685 0.0665         3
## 4 1         rLAP_i  0.210 0.725 0.0673         3
## 5 0         rHAN_i -0.401 0.608 0.0591         4
## 6 1         rHAN_i -0.164 0.607 0.0564         4
## 7 0         rLAN_i  0.303 0.642 0.0623         6
## 8 1         rLAN_i  0.191 0.763 0.0709         6
## # A tibble: 8 x 6
## # Groups:   cultureEA [2]
##   cultureEA AVI      mean    sd     se AVI_order
##   <fct>     <fct>   <dbl> <dbl>  <dbl>     <dbl>
## 1 0         iHAP_i  0.579 0.447 0.0434         1
## 2 1         iHAP_i  0.714 0.490 0.0455         1
## 3 0         iLAP_i  1.13  0.400 0.0388         3
## 4 1         iLAP_i  1.02  0.458 0.0425         3
## 5 0         iHAN_i -0.768 0.260 0.0252         4
## 6 1         iHAN_i -0.670 0.378 0.0351         4
## 7 0         iLAN_i -0.678 0.326 0.0317         6
## 8 1         iLAN_i -0.613 0.396 0.0368         6
```


Group Differences in Actual Affect
Ipsatized Scores
0.50
0.25
0.00
−0.25
−0.50
rHAP_i
rLAP_i
rHAN_i
rLAN_i
AVI
0
1



Analyses

```
# convert AVI data into long form
longavidata <- data %>%
  dplyr::select(
    subID, cultureEA, starts_with("exclude"),
    iHAP, iHAP_i, iLAP, iLAP_i,
    rHAP, rHAP_i, rLAP, rLAP_i) %>%
  pivot_longer(
    cols = `iHAP`:`iLAP_i`,
    names_to = "IdealAffect",
    values_to = "value")
```

```
# test if cultures differ on ideal affect (ipsatized scores)
idealAffectresult_i <-
  aov(value ~ IdealAffect*cultureEA + rHAP + rLAP + Error(subID/IdealAffect),
    data = longavidata %>%
      dplyr::filter(IdealAffect == "iHAP_i" |
                    IdealAffect == "iLAP_i"))
summary(idealAffectresult_i)
```

```
##
## Error: subID
```

```
##            Df Sum Sq Mean Sq F value Pr(>F)
## cultureEA   1   0.02 0.01884   0.097  0.756
## rHAP        1   0.03 0.03107   0.160  0.690
## rLAP        1   0.05 0.04606   0.236  0.627
## Residuals 218  42.46 0.19476
##
## Error: subID:IdealAffect
##                        Df Sum Sq Mean Sq F value  Pr(>F)
## IdealAffect             1  19.94  19.944  93.266 < 2e-16 ***
## IdealAffect:cultureEA   1   1.65   1.654   7.734 0.00589 **
## Residuals             220  47.04   0.214
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# pairwise tests (ipsatized scores)
idealHAPresult_i <-
  lm(value ~ cultureEA + rHAP,
     data = longavidata %>% dplyr::filter(IdealAffect == "iHAP_i"))
summary(idealHAPresult_i)
```

```
##
## Call:
## lm(formula = value ~ cultureEA + rHAP, data = longavidata %>%
##     dplyr::filter(IdealAffect == "iHAP_i"))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.66299 -0.24628  0.01261  0.33384  1.09651
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.26987    0.11268   2.395  0.01746 *
## cultureEA    0.12887    0.06213   2.074  0.03922 *
## rHAP         0.11591    0.03880   2.987  0.00314 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4621 on 219 degrees of freedom
## Multiple R-squared:  0.05874,    Adjusted R-squared:  0.05014
## F-statistic: 6.833 on 2 and 219 DF,  p-value: 0.001322
```

```r
idealLAPresult_i <-
  lm(value ~ cultureEA + rLAP,
     data = longavidata %>% dplyr::filter(IdealAffect == "iLAP_i"))
summary(idealLAPresult_i)
```

```
##
## Call:
## lm(formula = value ~ cultureEA + rLAP, data = longavidata %>%
##     dplyr::filter(IdealAffect == "iLAP_i"))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

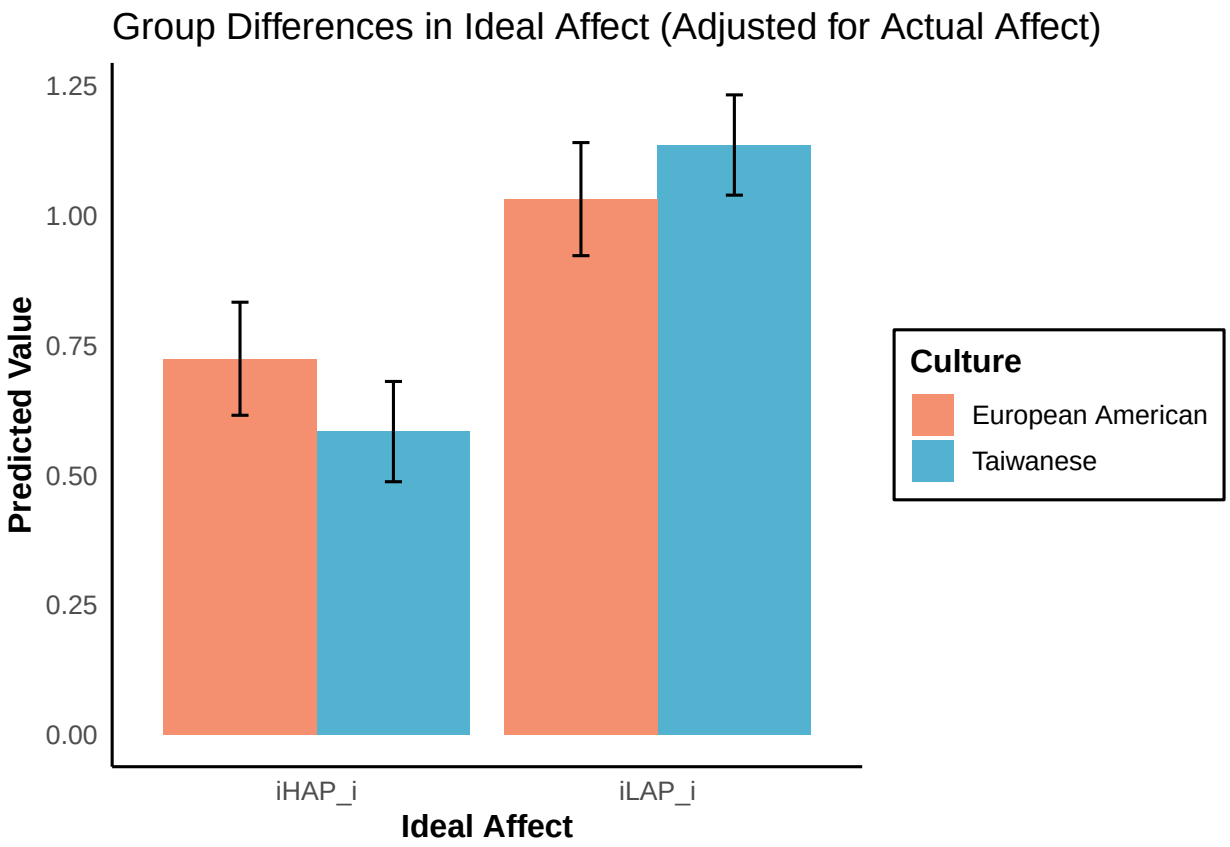
```
## -1.5932 -0.2004  0.1054  0.2813  0.7385
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.94715    0.11510   8.229 1.68e-14 ***
## cultureEA   -0.08999    0.05878  -1.531   0.1272
## rLAP         0.05942    0.03483   1.706   0.0894 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4294 on 219 degrees of freedom
## Multiple R-squared:  0.02878,    Adjusted R-squared:  0.01991
## F-statistic: 3.245 on 2 and 219 DF,  p-value: 0.04087
```

```r
emm <- emmeans(
  idealAffectresult_i,
  ~ IdealAffect * cultureEA
)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## Warning in (mth$objs[[1]])(object, trms, xlev, grid, ...): Some predictors are
## correlated with the intercept - results may be very biased
```

```r
plot_df <- as.data.frame(emm) %>%
  dplyr::mutate(
    Culture = if_else(cultureEA == 1, "European American", "Taiwanese") %>%
      as.factor()
  )

ggplot(plot_df %>%
         filter(grepl("_i", IdealAffect)),  # keep only the relevant IdealAffect levels
       aes(x = IdealAffect, y = emmean, fill = Culture)) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = emmean - SE, ymax = emmean + SE),
                position = position_dodge(0.9), width = 0.1) +
  scale_fill_manual(values = c(
    "European American"="#F49070",
    "Taiwanese"="#52B2CF")) +
  xlab("Ideal Affect") +
  ylab("Predicted Value") +
  ggtitle("Group Differences in Ideal Affect (Adjusted for Actual Affect)")
```


Group Differences in Ideal Affect (Adjusted for Actual Affect)
1.25
1.00
0.75
0.50
0.25
0.00
Predicted Value
iHAP_i
iLAP_i
Ideal Affect
Culture
European American
Taiwanese

Does Induced Mood Influence Giving Towards Excited, Calm, and Neutral Responders Differently Across Cultures?

Admin before analyses

```
# convert giving data into long form
longGivingdata_i <- data %>%
  dplyr::select(
    subID, cultureEA, order,
    starts_with("raw") & contains("_i"),
    starts_with("exclude"),
    age
    ) %>%
  pivot_longer(
    cols = `raw1_i`:`raw48_i`,
    names_to = "trial",
    values_to = "response_i") %>%
  dplyr::mutate(trial = as.numeric(substr(trial, start = 4, stop = str_length(trial)-2)))
```

```
# add info on incidental affect (mood), face gender, face race, and face emotion for each trial.
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    mood = levelBreakdown$mood[trial],
    faceGender = levelBreakdown$faceGender[trial],
    faceRace = levelBreakdown$faceRace[trial],
    faceEmotion = levelBreakdown$faceEmotion[trial])

# make variables factors
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    culture = ifelse(cultureEA == 1, "EA", "TN") %>% as.factor(),
    mood = factor(mood, levels = c("HAP", "HAN","Neutral")),
    faceRace = factor(faceRace, levels = c("White", "Asian")),
    faceGender = factor(faceGender, levels = c("Male", "Female")),
    faceEmotion = factor(faceEmotion, levels = c(
      "Excited", "Calm", "Neutral", "Negative")),
    subID = factor(subID, unique(subID)))

longGivingdata_i = longGivingdata_i %>% dplyr::filter(!is.na(response_i))
```

Sample size

```r
print("n_total:"); longGivingdata_i %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 187
```

```r
print("n_tw:"); longGivingdata_i %>%
  dplyr::filter(culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 96
```

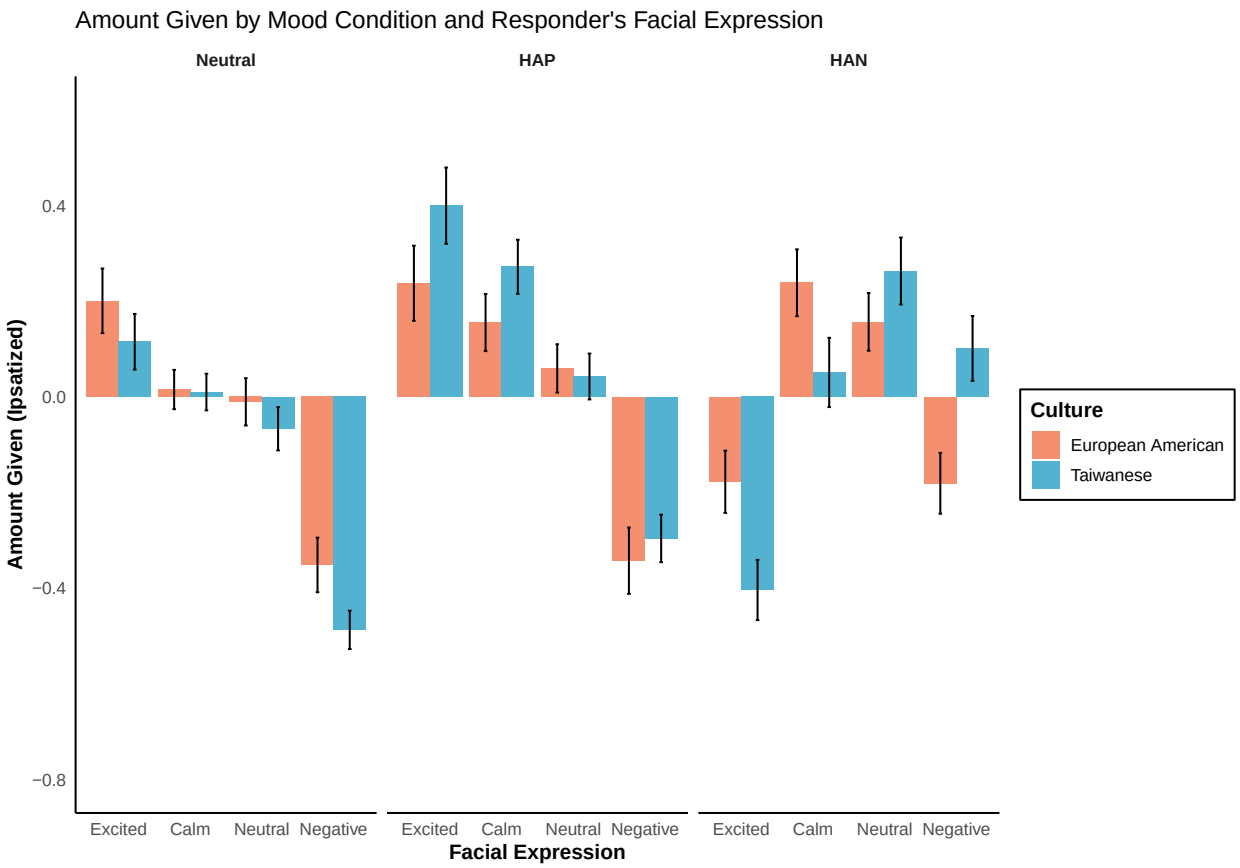
```r
print("n_ea:"); longGivingdata_i %>%
  dplyr::filter(culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 91
```

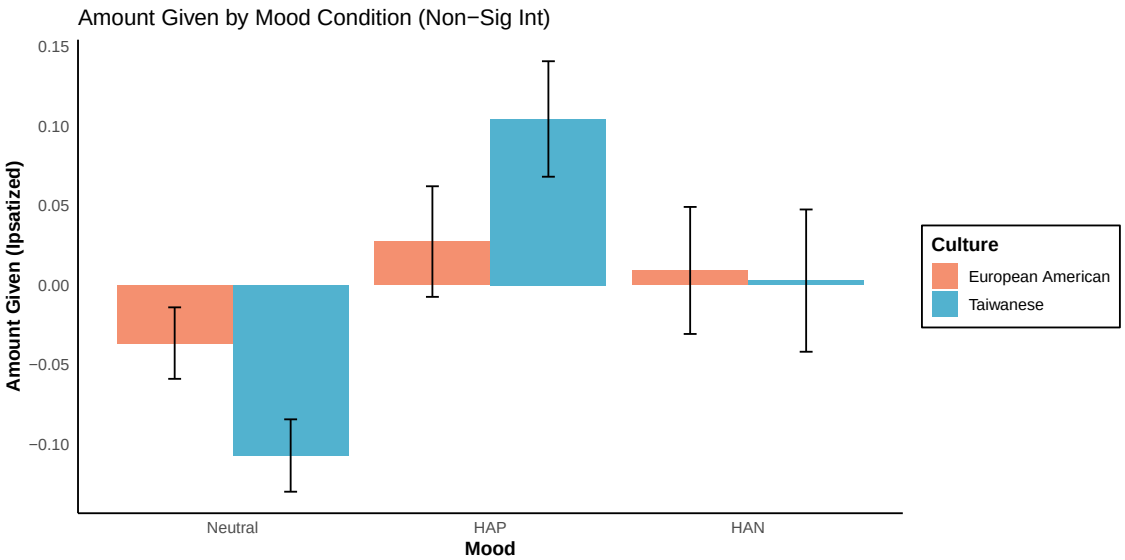
Visualizations

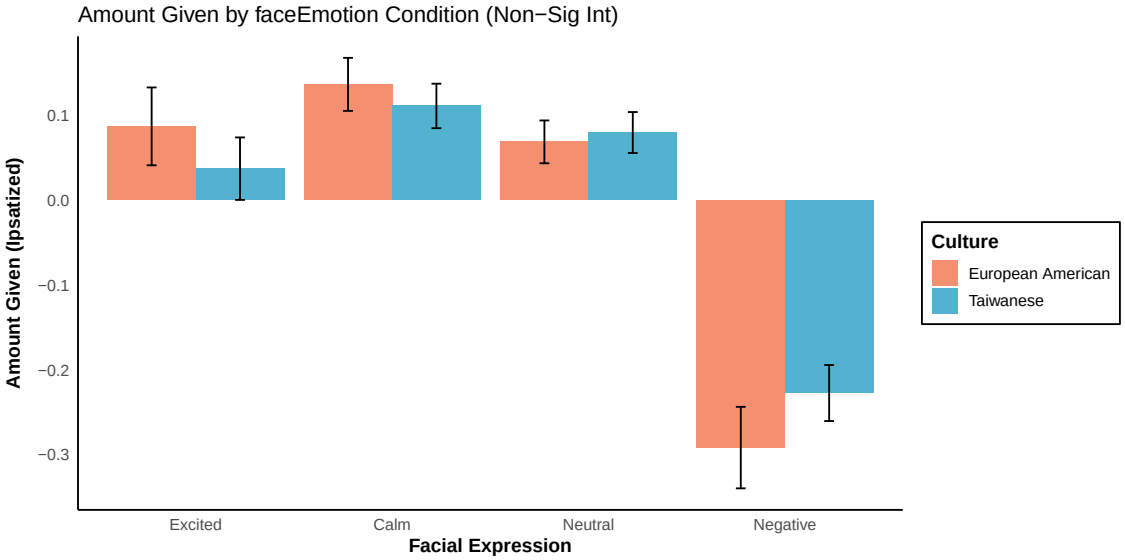
```
longGivingdata_i2 <- longGivingdata_i %>% dplyr::mutate(
  culture = if_else(culture == "EA", "European American", "Taiwanese")) %>%
  dplyr::rename(Culture = culture) %>%
  dplyr::mutate(
    mood = factor(mood,
                  labels = c("Neutral", "HAP", "HAN"),
                  levels = c("Neutral", "HAP", "HAN")))
ggplot(
  longGivingdata_i2 %>%
    group_by(subID, mood, faceEmotion, Culture) %>%
    summarise(response_i = mean(response_i, na.rm = T)) %>%
    group_by(faceEmotion, mood, Culture) %>%
    summarise(mean = mean(response_i, na.rm = T),
              SD = sd(response_i, na.rm = T),
              SE = sd(response_i, na.rm = T)/sqrt(n())),
  aes(faceEmotion, mean, fill = Culture)) +
  facet_grid(~mood)+
  scale_fill_manual(
    values = c("European American"="#F49070", "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1)+
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ylim(c(-0.8, 0.6)) +
  ggtitle("Amount Given by Mood Condition and Responder's Facial Expression")
```

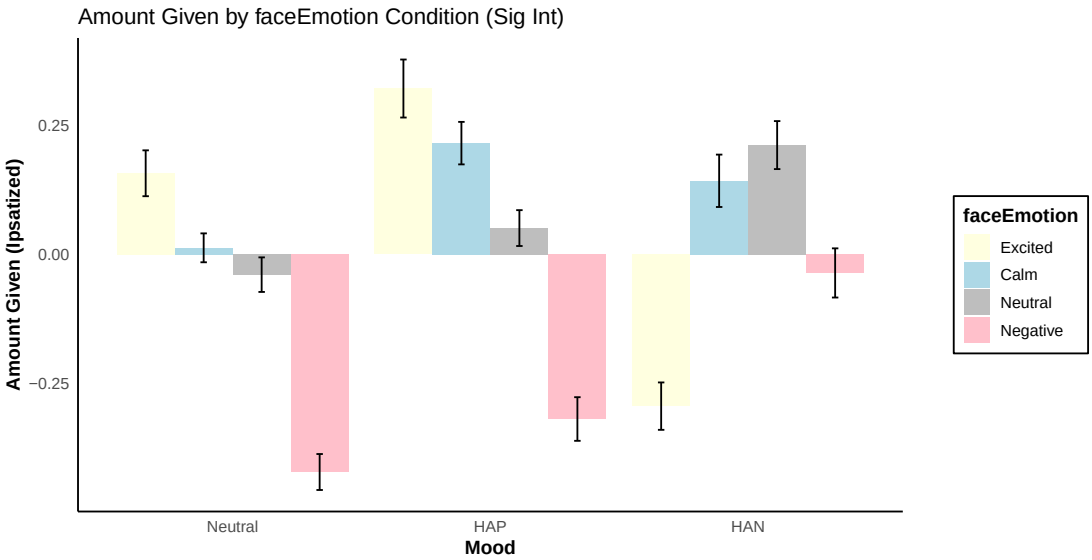

Amount Given by Mood Condition and Responder's Facial Expression
Neutral
HAP
HAN
Amount Given (Ipsatized)
0.4
0.0
−0.4
−0.8
Excited
Calm
Neutral
Negative
Facial Expression
Culture
European American
Taiwanese

```r
p1 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, mood, Culture) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(mood, Culture) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by Mood Condition (Non-Sig Int)")

p2 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, faceEmotion, Culture) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(faceEmotion, Culture) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(faceEmotion, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition (Non-Sig Int)")

p3 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, faceEmotion, mood) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(faceEmotion, mood) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = faceEmotion)) +
  scale_fill_manual(values =
                      c("Neutral"="grey",
                        "Excited"="lightyellow",
                        "Calm"="lightblue",
                        "Negative"="pink")) +
  geom_bar(stat='identity', position='dodge') +
```

```r
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition (Sig Int)")

p1 + p2 + p3 +
  patchwork::plot_layout(
    ncol = 1
  )
```


Amount Given by Mood Condition (Non−Sig Int)
Amount Given (Ipsatized)
0.15
0.10
0.05
0.00
−0.05
−0.10
Neutral
HAP
HAN
Mood
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition (Non−Sig Int)
Amount Given (Ipsatized)
0.1
0.0
−0.1
−0.2
−0.3
Excited
Calm
Neutral
Negative
Facial Expression
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition (Sig Int)
Amount Given (Ipsatized)
0.25
0.00
−0.25
Neutral
HAP
HAN
Mood
faceEmotion
Excited
Calm
Neutral
Negative

Analyses

```r
aov(
  formula = response_i ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i) %>% summary()
```

```
##
## Error: subID
##            Df    Sum Sq   Mean Sq F value Pr(>F)
## culture     1 1.400e-28 1.394e-28   0.298  0.586
## Residuals 185 8.655e-26 4.678e-28
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value  Pr(>F)
## mood           2   29.4  14.695   5.466 0.00458 **
## mood:culture   2    8.2   4.108   1.528 0.21832
## Residuals    370  994.8   2.689
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value Pr(>F)
## faceEmotion           3  205.0   68.32  38.008 <2e-16 ***
## culture:faceEmotion   3    4.1    1.38   0.767  0.513
## Residuals           555  997.6    1.80
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                            Df Sum Sq Mean Sq F value  Pr(>F)
## mood:faceEmotion            6  221.4   36.90  40.122 < 2e-16 ***
## mood:culture:faceEmotion    6   34.1    5.68   6.174 2.2e-06 ***
## Residuals                1110 1021.0    0.92
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 6732   5273  0.7833
```

Pairwise

```
pairwise <- aov(
  formula = response_i ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i)
```

Main effect mood

```r
# main effect mood
emm_3way_mood <- emmeans(pairwise, ~ mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_mood) # p values
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  HAP - HAN       0.0599 0.0424 370   1.413  0.3351
##  HAP - Neutral   0.1378 0.0424 370   3.250  0.0036
##  HAN - Neutral   0.0779 0.0424 370   1.836  0.1592
##
## Results are averaged over the levels of: culture, faceEmotion
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_mood)) # confidence intervals
```

Main effect faceEmotion

```r
# main effect faceEmotion
emm_3way_faceEmotion <- emmeans(pairwise, ~ faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotion) # p values
```

```
##  contrast           estimate   SE  df t.ratio p.value
##  Excited - Calm      -0.0617 0.04 555  -1.541  0.4139
##  Excited - Neutral   -0.0122 0.04 555  -0.303  0.9903
##  Excited - Negative   0.3214 0.04 555   8.027  <.0001
##  Calm - Neutral       0.0495 0.04 555   1.237  0.6035
##  Calm - Negative      0.3831 0.04 555   9.567  <.0001
##  Neutral - Negative   0.3335 0.04 555   8.330  <.0001
##
## Results are averaged over the levels of: mood, culture
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```r
#confint(pairs(emm_3way_faceEmotion)) # confidence intervals
```

Mood:faceEmotion

```r
emm_3way_faceEmotionMood <- emmeans(pairwise, ~ faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionMood) # p values
```

```
## mood = HAP:
##  contrast           estimate    SE   df t.ratio p.value
##  Excited - Calm       0.1050 0.057 1491   1.844  0.2531
##  Excited - Neutral    0.2676 0.057 1491   4.698  <.0001
##  Excited - Negative   0.6378 0.057 1491  11.198  <.0001
##  Calm - Neutral       0.1625 0.057 1491   2.854  0.0227
##  Calm - Negative      0.5328 0.057 1491   9.354  <.0001
##  Neutral - Negative   0.3702 0.057 1491   6.500  <.0001
##
## mood = HAN:
##  contrast           estimate    SE   df t.ratio p.value
##  Excited - Calm      -0.4354 0.057 1491  -7.645  <.0001
##  Excited - Neutral   -0.5006 0.057 1491  -8.789  <.0001
##  Excited - Negative  -0.2509 0.057 1491  -4.405  0.0001
##  Calm - Neutral      -0.0652 0.057 1491  -1.144  0.6623
##  Calm - Negative      0.1845 0.057 1491   3.240  0.0067
##  Neutral - Negative   0.2497 0.057 1491   4.384  0.0001
##
## mood = Neutral:
##  contrast           estimate    SE   df t.ratio p.value
##  Excited - Calm       0.1453 0.057 1491   2.552  0.0527
##  Excited - Neutral    0.1966 0.057 1491   3.451  0.0032
##  Excited - Negative   0.5773 0.057 1491  10.135  <.0001
##  Calm - Neutral       0.0512 0.057 1491   0.899  0.8052
##  Calm - Negative      0.4319 0.057 1491   7.584  <.0001
##  Neutral - Negative   0.3807 0.057 1491   6.684  <.0001
##
## Results are averaged over the levels of: culture
## P value adjustment: tukey method for comparing a family of 4 estimates
```

```r
#confint(pairs(emm_3way_faceEmotionMood)) # confidence intervals
```

Mood:culture:faceEmotion

```r
emm_3way_bymood <- emmeans(pairwise, ~ culture * faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_bymood) # p values
```

```
## mood = HAP:
##  contrast                  estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited   -0.16229 0.0852 1679  -1.904  0.5482
##  EA Excited - EA Calm       0.08220 0.0816 1491   1.007  0.9735
##  EA Excited - TN Calm      -0.03443 0.0852 1679  -0.404  0.9999
##  EA Excited - EA Neutral    0.17795 0.0816 1491   2.180  0.3642
##  EA Excited - TN Neutral    0.19489 0.0852 1679   2.287  0.3015
##  EA Excited - EA Negative   0.57992 0.0816 1491   7.105  <.0001
##  EA Excited - TN Negative   0.53339 0.0852 1679   6.258  <.0001
##  TN Excited - EA Calm       0.24449 0.0852 1679   2.869  0.0797
##  TN Excited - TN Calm       0.12786 0.0795 1491   1.609  0.7450
##  TN Excited - EA Neutral    0.34024 0.0852 1679   3.992  0.0018
##  TN Excited - TN Neutral    0.35718 0.0795 1491   4.495  0.0002
##  TN Excited - EA Negative   0.74221 0.0852 1679   8.709  <.0001
##  TN Excited - TN Negative   0.69568 0.0795 1491   8.755  <.0001
##  EA Calm - TN Calm         -0.11663 0.0852 1679  -1.368  0.8716
##  EA Calm - EA Neutral       0.09575 0.0816 1491   1.173  0.9395
##  EA Calm - TN Neutral       0.11269 0.0852 1679   1.322  0.8906
##  EA Calm - EA Negative      0.49772 0.0816 1491   6.098  <.0001
##  EA Calm - TN Negative      0.45119 0.0852 1679   5.294  <.0001
##  TN Calm - EA Neutral       0.21238 0.0852 1679   2.492  0.1995
##  TN Calm - TN Neutral       0.22931 0.0795 1491   2.886  0.0762
##  TN Calm - EA Negative      0.61435 0.0852 1679   7.208  <.0001
##  TN Calm - TN Negative      0.56782 0.0795 1491   7.146  <.0001
##  EA Neutral - TN Neutral    0.01693 0.0852 1679   0.199  1.0000
##  EA Neutral - EA Negative   0.40197 0.0816 1491   4.925  <.0001
##  EA Neutral - TN Negative   0.35544 0.0852 1679   4.170  0.0008
##  TN Neutral - EA Negative   0.38503 0.0852 1679   4.518  0.0002
##  TN Neutral - TN Negative   0.33851 0.0795 1491   4.260  0.0006
##  EA Negative - TN Negative -0.04653 0.0852 1679  -0.546  0.9994
##
## mood = HAN:
##  contrast                  estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited    0.22623 0.0852 1679   2.654  0.1376
##  EA Excited - EA Calm      -0.41605 0.0816 1491  -5.098  <.0001
##  EA Excited - TN Calm      -0.22855 0.0852 1679  -2.682  0.1289
##  EA Excited - EA Neutral   -0.33436 0.0816 1491  -4.097  0.0011
##  EA Excited - TN Neutral   -0.44055 0.0852 1679  -5.169  <.0001
##  EA Excited - EA Negative   0.00321 0.0816 1491   0.039  1.0000
##  EA Excited - TN Negative  -0.27875 0.0852 1679  -3.271  0.0242
##  TN Excited - EA Calm      -0.64228 0.0852 1679  -7.536  <.0001
##  TN Excited - TN Calm      -0.45478 0.0795 1491  -5.723  <.0001
##  TN Excited - EA Neutral   -0.56059 0.0852 1679  -6.578  <.0001
##  TN Excited - TN Neutral   -0.66678 0.0795 1491  -8.391  <.0001
##  TN Excited - EA Negative  -0.22302 0.0852 1679  -2.617  0.1505
```

```
##  TN Excited - TN Negative  -0.50498 0.0795 1491  -6.355  <.0001
##  EA Calm - TN Calm          0.18750 0.0852 1679   2.200  0.3521
##  EA Calm - EA Neutral       0.08169 0.0816 1491   1.001  0.9744
##  EA Calm - TN Neutral      -0.02450 0.0852 1679  -0.287  1.0000
##  EA Calm - EA Negative      0.41926 0.0816 1491   5.137  <.0001
##  EA Calm - TN Negative      0.13730 0.0852 1679   1.611  0.7438
##  TN Calm - EA Neutral      -0.10581 0.0852 1679  -1.242  0.9194
##  TN Calm - TN Neutral      -0.21200 0.0795 1491  -2.668  0.1333
##  TN Calm - EA Negative      0.23176 0.0852 1679   2.719  0.1174
##  TN Calm - TN Negative     -0.05020 0.0795 1491  -0.632  0.9984
##  EA Neutral - TN Neutral   -0.10619 0.0852 1679  -1.246  0.9180
##  EA Neutral - EA Negative   0.33757 0.0816 1491   4.136  0.0010
##  EA Neutral - TN Negative   0.05561 0.0852 1679   0.652  0.9981
##  TN Neutral - EA Negative   0.44376 0.0852 1679   5.207  <.0001
##  TN Neutral - TN Negative   0.16180 0.0795 1491   2.036  0.4576
##  EA Negative - TN Negative -0.28196 0.0852 1679  -3.308  0.0215
##
## mood = Neutral:
##  contrast                  estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited    0.08558 0.0852 1679   1.004  0.9739
##  EA Excited - EA Calm       0.18547 0.0816 1491   2.272  0.3096
##  EA Excited - TN Calm       0.19077 0.0852 1679   2.238  0.3292
##  EA Excited - EA Neutral    0.21118 0.0816 1491   2.587  0.1612
##  EA Excited - TN Neutral    0.26750 0.0852 1679   3.139  0.0367
##  EA Excited - EA Negative   0.55219 0.0816 1491   6.766  <.0001
##  EA Excited - TN Negative   0.68792 0.0852 1679   8.072  <.0001
##  TN Excited - EA Calm       0.09989 0.0852 1679   1.172  0.9398
##  TN Excited - TN Calm       0.10519 0.0795 1491   1.324  0.8900
##  TN Excited - EA Neutral    0.12560 0.0852 1679   1.474  0.8216
##  TN Excited - TN Neutral    0.18192 0.0795 1491   2.289  0.3000
##  TN Excited - EA Negative   0.46661 0.0852 1679   5.475  <.0001
##  TN Excited - TN Negative   0.60234 0.0795 1491   7.580  <.0001
##  EA Calm - TN Calm          0.00530 0.0852 1679   0.062  1.0000
##  EA Calm - EA Neutral       0.02571 0.0816 1491   0.315  1.0000
##  EA Calm - TN Neutral       0.08203 0.0852 1679   0.963  0.9795
##  EA Calm - EA Negative      0.36672 0.0816 1491   4.493  0.0002
##  EA Calm - TN Negative      0.50245 0.0852 1679   5.895  <.0001
##  TN Calm - EA Neutral       0.02041 0.0852 1679   0.240  1.0000
##  TN Calm - TN Neutral       0.07674 0.0795 1491   0.966  0.9791
##  TN Calm - EA Negative      0.36143 0.0852 1679   4.241  0.0006
##  TN Calm - TN Negative      0.49715 0.0795 1491   6.256  <.0001
##  EA Neutral - TN Neutral    0.05632 0.0852 1679   0.661  0.9979
##  EA Neutral - EA Negative   0.34101 0.0816 1491   4.178  0.0008
##  EA Neutral - TN Negative   0.47674 0.0852 1679   5.594  <.0001
##  TN Neutral - EA Negative   0.28469 0.0852 1679   3.340  0.0193
##  TN Neutral - TN Negative   0.42042 0.0795 1491   5.291  <.0001
##  EA Negative - TN Negative  0.13573 0.0852 1679   1.593  0.7550
##
## P value adjustment: tukey method for comparing a family of 8 estimates
```

```
#confint(pairs(emm_3way_bymood)) # confidence intervals
```

```r
emm_3way_byfaceEmotion <- emmeans(pairwise, ~ culture * mood | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_byfaceEmotion) # p values
```

```
## faceEmotion = Excited:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP         -0.16229 0.0852 1679  -1.904  0.3998
##  EA HAP - EA HAN          0.41499 0.0865 1124   4.797  <.0001
##  EA HAP - TN HAN          0.64122 0.0852 1679   7.524  <.0001
##  EA HAP - EA Neutral      0.03669 0.0865 1124   0.424  0.9983
##  EA HAP - TN Neutral      0.12227 0.0852 1679   1.435  0.7058
##  TN HAP - EA HAN          0.57728 0.0852 1679   6.773  <.0001
##  TN HAP - TN HAN          0.80351 0.0842 1124   9.540  <.0001
##  TN HAP - EA Neutral      0.19898 0.0852 1679   2.335  0.1807
##  TN HAP - TN Neutral      0.28456 0.0842 1124   3.379  0.0098
##  EA HAN - TN HAN          0.22623 0.0852 1679   2.654  0.0852
##  EA HAN - EA Neutral     -0.37830 0.0865 1124  -4.373  0.0002
##  EA HAN - TN Neutral     -0.29272 0.0852 1679  -3.435  0.0080
##  TN HAN - EA Neutral     -0.60453 0.0852 1679  -7.093  <.0001
##  TN HAN - TN Neutral     -0.51895 0.0842 1124  -6.162  <.0001
##  EA Neutral - TN Neutral  0.08558 0.0852 1679   1.004  0.9167
##
## faceEmotion = Calm:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP         -0.11663 0.0852 1679  -1.368  0.7460
##  EA HAP - EA HAN         -0.08326 0.0865 1124  -0.962  0.9296
##  EA HAP - TN HAN          0.10424 0.0852 1679   1.223  0.8257
##  EA HAP - EA Neutral      0.13996 0.0865 1124   1.618  0.5868
##  EA HAP - TN Neutral      0.14526 0.0852 1679   1.704  0.5290
##  TN HAP - EA HAN          0.03337 0.0852 1679   0.391  0.9988
##  TN HAP - TN HAN          0.22087 0.0842 1124   2.622  0.0927
##  TN HAP - EA Neutral      0.25659 0.0852 1679   3.011  0.0316
##  TN HAP - TN Neutral      0.26189 0.0842 1124   3.109  0.0235
##  EA HAN - TN HAN          0.18750 0.0852 1679   2.200  0.2382
##  EA HAN - EA Neutral      0.22322 0.0865 1124   2.580  0.1029
##  EA HAN - TN Neutral      0.22852 0.0852 1679   2.681  0.0795
##  TN HAN - EA Neutral      0.03572 0.0852 1679   0.419  0.9984
##  TN HAN - TN Neutral      0.04102 0.0842 1124   0.487  0.9966
##  EA Neutral - TN Neutral  0.00530 0.0852 1679   0.062  1.0000
##
## faceEmotion = Neutral:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.01693 0.0852 1679   0.199  1.0000
##  EA HAP - EA HAN         -0.09733 0.0865 1124  -1.125  0.8710
##  EA HAP - TN HAN         -0.20351 0.0852 1679  -2.388  0.1610
##  EA HAP - EA Neutral      0.06992 0.0865 1124   0.808  0.9661
##  EA HAP - TN Neutral      0.12624 0.0852 1679   1.481  0.6764
##  TN HAP - EA HAN         -0.11426 0.0852 1679  -1.341  0.7622
##  TN HAP - TN HAN         -0.22045 0.0842 1124  -2.617  0.0939
##  TN HAP - EA Neutral      0.05298 0.0852 1679   0.622  0.9895
```

```
##  TN HAP - TN Neutral      0.10931 0.0842 1124   1.298  0.7864
##  EA HAN - TN HAN         -0.10619 0.0852 1679  -1.246  0.8140
##  EA HAN - EA Neutral      0.16724 0.0865 1124   1.933  0.3822
##  EA HAN - TN Neutral      0.22357 0.0852 1679   2.623  0.0923
##  TN HAN - EA Neutral      0.27343 0.0852 1679   3.208  0.0171
##  TN HAN - TN Neutral      0.32975 0.0842 1124   3.915  0.0013
##  EA Neutral - TN Neutral  0.05632 0.0852 1679   0.661  0.9861
##
## faceEmotion = Negative:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP         -0.04653 0.0852 1679  -0.546  0.9942
##  EA HAP - EA HAN         -0.16172 0.0865 1124  -1.869  0.4217
##  EA HAP - TN HAN         -0.44368 0.0852 1679  -5.206  <.0001
##  EA HAP - EA Neutral      0.00896 0.0865 1124   0.104  1.0000
##  EA HAP - TN Neutral      0.14469 0.0852 1679   1.698  0.5335
##  TN HAP - EA HAN         -0.11519 0.0852 1679  -1.352  0.7559
##  TN HAP - TN HAN         -0.39715 0.0842 1124  -4.715  <.0001
##  TN HAP - EA Neutral      0.05549 0.0852 1679   0.651  0.9870
##  TN HAP - TN Neutral      0.19122 0.0842 1124   2.270  0.2071
##  EA HAN - TN HAN         -0.28196 0.0852 1679  -3.308  0.0123
##  EA HAN - EA Neutral      0.17068 0.0865 1124   1.973  0.3585
##  EA HAN - TN Neutral      0.30641 0.0852 1679   3.595  0.0045
##  TN HAN - EA Neutral      0.45264 0.0852 1679   5.311  <.0001
##  TN HAN - TN Neutral      0.58837 0.0842 1124   6.986  <.0001
##  EA Neutral - TN Neutral  0.13573 0.0852 1679   1.593  0.6036
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
#confint(pairs(emm_3way_byfaceEmotion)) # confidence intervals
```